

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2436

Stuttgart Ricuskey 100 kV
September 20, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2436. The Interconnection Customer proposes a 46.5 MW battery energy storage facility interconnecting at the Stuttgart Ricuskey 100 kV substation in the Illinois area. The request was received on July 28, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,264,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Flambeau Solar Partners – Pampanos 100 kV Cir. 1	Flambeau Solar Partners – Pampanos 100 kV #2	115.0	143	16.6
Flambeau Solar Partners – Pampanos 100 kV Ckt 2	Flambeau Solar Partners – Pampanos Substation 100 kV	115.0	143	16.6
Big Oaks Substation – Cypress Creek 100 kV	Jones Chapel – Dekalb 500 kV	107.2	500	5.2
Big Oaks – Laureles 100 kV	Dealb – Jones Chapel 500 kV	106.2	531	5.2
Waynedale – Montpelier Electric Generating Station 230 kV	Richland – Dekalb 500 kV line	105.7	1,919	20.4
Ward Gulf – Laureles 100 kV	Dekalb – Jones Chapel 500 kV	105.2	562	5.2
Ege – Golden Gate 100 kV Line	Dekalb – Jones Chapel 500 kV line	104.6	398	5.2
Ege – Cuba City 100 kV	Dekalb – Jones Chapel 500 kV	103.9	414	5.2
Minnehaa – Sidney A Murray Jr Hydroelectric 230 kV line	Richland – Dekalb 500 kV	103.5	2,041	16.0
Golden Gate – Laotto Substation 100 kV Line	Deklb – Jones Chapel 500 kV	103.3	385	5.2
Pickford – Herald-Twin Cities 230 kV	Beaver – Trego 500 kV	102.9	2,114	9.4
Cuba City – Farmersville 100 kV line	Dekalb – Jones Chapel 500 kV	102.5	448	5.2
Post Rock – Newport Tie 230 kV	Trego – Beaver 500 kV L	102.3	2,963	12.8
Post Rock – Myrtle Point 230 kV	Trego – Beaver 500 kV	102.3	2,995	12.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Canal Rd – Weedpatch 230 kV	Beaver – Trego 500 kV	102.2	1,782	7.4
Rex Brown – Kendallville 100 kV	Dekalb – Jnes Chapel 500 kV	102.1	487	5.2
Rex Brown – Montibello 100 kV	Dekalb – Jones Chapel 500 kV	102.0	506	5.2
Ligonier – Montibello 100 kV	Deklb – Jones Chapel 500 kV	101.9	519	5.2
Farmersville – Kendallville 100 kV L	Dekalb – Jones Chapel 500 kV L	101.8	475	5.2
Monfort Substation – Wabash 230 kV	Richland – Dekalb 500 kV L	100.8	1,847	17.7
Ligonier – Cypress Crk 100 kV	Dekalb – Jones Chapel 500 kV	100.2	529	5.2
Station S Van Nuys 500/230kV Transformer Bank	Richland – Troy Energy Llc 500 kV	100.1	1,015	7.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Flambeau Solar Partners – Pampanos 100 kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,332,000	30% (\$403,000)
Flambeau Solar Partners – Pampanos 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,116,000	30% (\$337,000)
Cypress Creek – Big Oaks 100 kV L	Reconductor line	\$2,369,000	0% (\$9,000)
Big Oaks – Laureles 100 kV	Reconductor line	\$734,000	0% (\$3,000)
Montpelier Electric Generating Sta – Waynedale 230 kV	Reconductor line	\$13,458,000	0% (\$55,000)
Ward Gulf – Laureles Substation 100 kV L	Reconductor line	\$1,978,000	0% (\$8,000)
Ege – Golden Gate 100 kV Line	Reconductor line	\$1,003,000	0% (\$4,000)
Ege – Cuba City 100 kV	Reconductor line	\$1,589,000	0% (\$6,000)
Minnehaa – Sidney A Murray Jr Hydroelectric 230 kV line	Reconductor line	\$10,462,000	0% (\$43,000)
Golden Gate – Laotto 100 kV	Reconductor line	\$2,749,000	0% (\$11,000)
Pickford – Herald-Twin Cities 230 kV	Reconductor line	\$14,656,000	0% (\$61,000)
Cuba City – Farmersville 100 kV line	Reconductor line	\$2,044,000	0% (\$8,000)
Post Rock – Newport Tie 230 kV	Reconductor line	\$9,672,000	0% (\$40,000)
Post Rock – Myrtle Pt 230 kV L	Reconductor line	\$11,986,000	0% (\$49,000)
Canal Rd – Weedpatch 230 kV	Reconductor line	\$6,044,000	0% (\$25,000)
Kendallville – Rex Brown 100 kV L	Reconductor line	\$1,487,000	0% (\$6,000)
Rex Brown – Montibello 100 kV	Reconductor line	\$1,905,000	0% (\$8,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ligonier – Montibello 100 kV	Reconductor line	\$2,324,000	0% (\$9,000)
Farmersville – Kendallville 100 kV L	Reconductor line	\$1,612,000	0% (\$6,000)
Monfort – Wabash Substation 230 kV	Reconductor line	\$11,146,000	0% (\$46,000)
Ligonier – Cypress Crk 100 kV	Reconductor line	\$2,332,000	0% (\$9,000)
Sta S Van Nuys 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$28,461,000	0% (\$118,000)
	Total allocated to project		\$1,264,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.